

Lab Experiments
Set:6-10
Date:20-07-2026

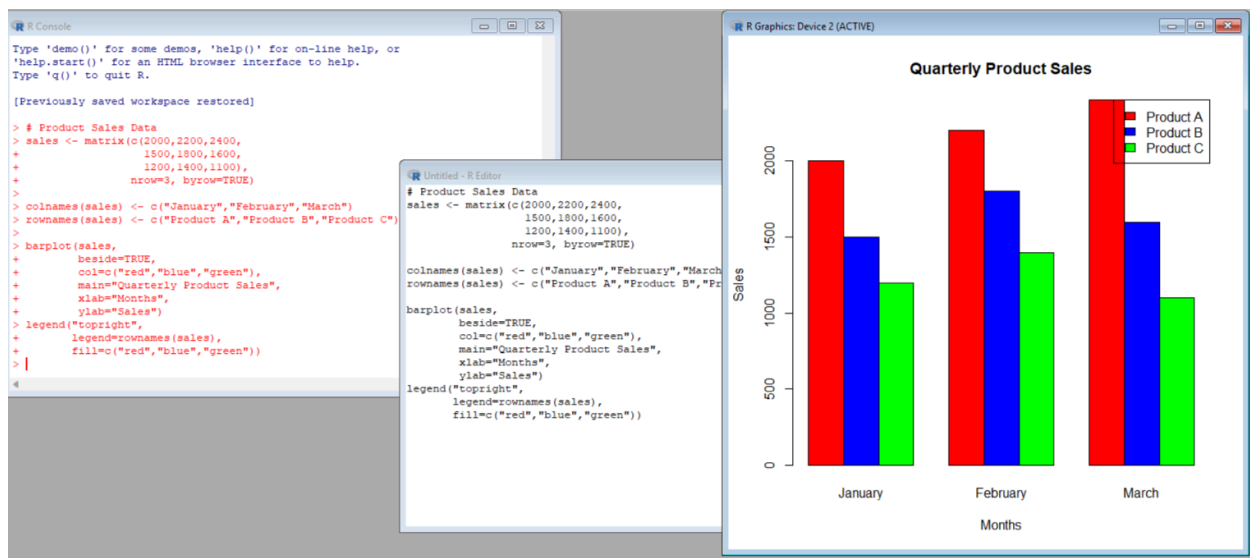
Set 6.Product Sales Analysis

1. Grouped Bar Chart

Code:

```
# Product Sales Data
sales <- matrix(c(2000,2200,2400,
                 1500,1800,1600,
                 1200,1400,1100),
               nrow=3, byrow=TRUE)
colnames(sales) <- c("January", "February", "March")
rownames(sales) <- c("Product A", "Product B", "Product C")
barplot(sales,
        beside=TRUE,
        col=c("red", "blue", "green"),
        main="Quarterly Product Sales",
        xlab="Months",
        ylab="Sales")
legend("topright",
      legend=rownames(sales),
      fill=c("red", "blue", "green"))
```

Output:

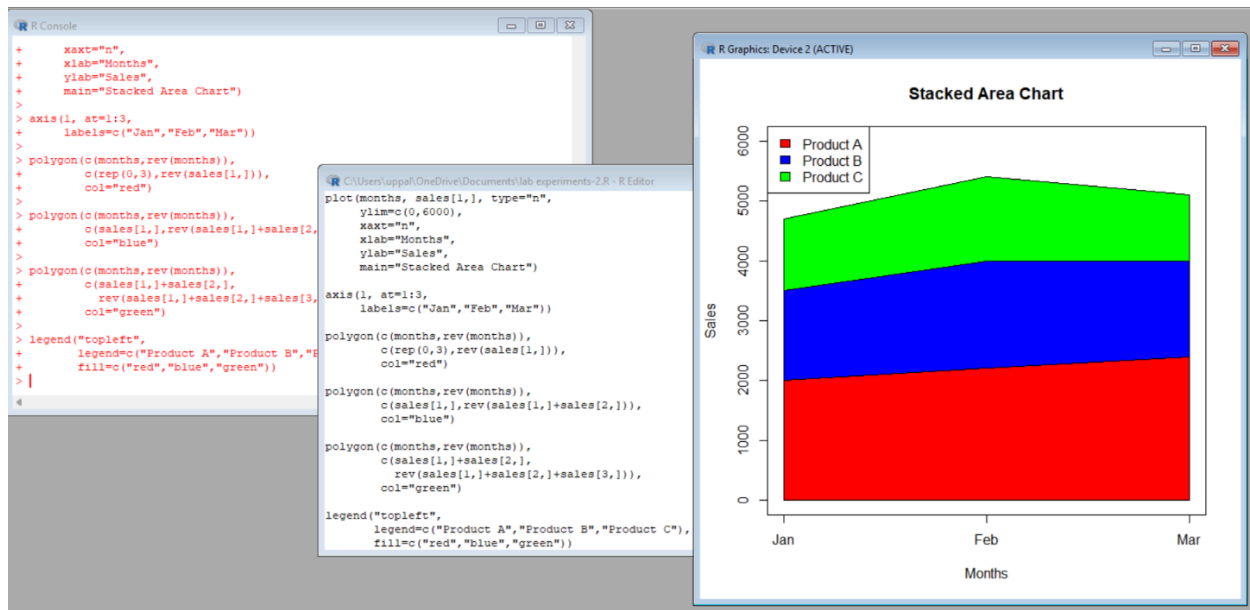


2. Stacked Area Chart

Code:

```
months <- 1:3
plot(months, sales[1,], type="n",
     ylim=c(0,6000),
     xaxt="n",
     xlab="Months",
     ylab="Sales",
     main="Stacked Area Chart")
axis(1, at=1:3,
     labels=c("Jan", "Feb", "Mar"))
polygon(c(months, rev(months)),
       c(rep(0,3), rev(sales[1,])),
       col="red")
polygon(c(months, rev(months)),
       c(sales[1,], rev(sales[1,]+sales[2,])),
       col="blue")
polygon(c(months, rev(months)),
       c(sales[1,]+sales[2,],
         rev(sales[1,]+sales[2,]+sales[3,])),
       col="green")
legend("topleft",
      legend=c("Product A", "Product B", "Product C"),
      fill=c("red", "blue", "green"))
```

Output:

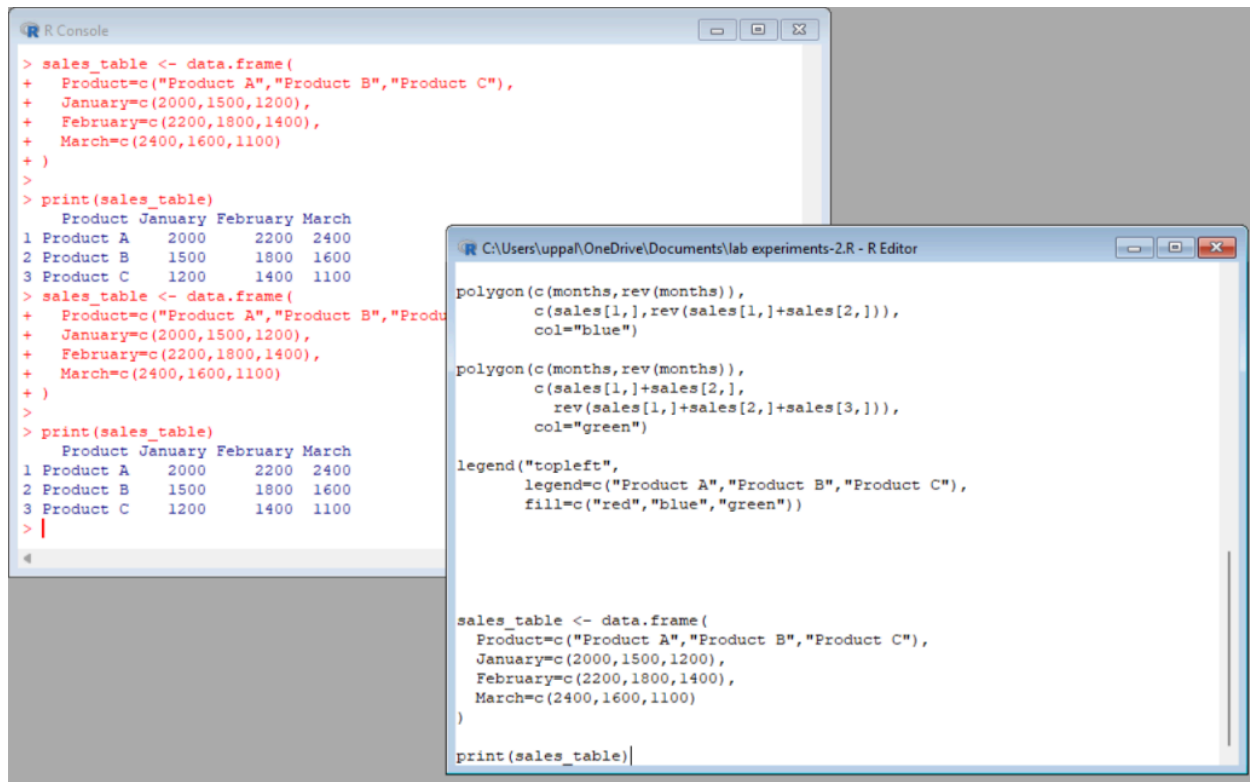


3. Monthly Sales Table

Code:

```
sales_table <- data.frame(  
  Product=c("Product A","Product B","Product C"),  
  January=c(2000,1500,1200),  
  February=c(2200,1800,1400),  
  March=c(2400,1600,1100)  
)  
print(sales_table)
```

Output:



4. Tableau Dashboard

Code:

```
library(shiny)  
sales <- matrix(c(2000,2200,2400,  
  1500,1800,1600,  
  1200,1400,1100),  
  nrow=3, byrow=TRUE)  
colnames(sales) <- c("January","February","March")  
rownames(sales) <- c("Product A","Product B","Product C")  
sales_table <- data.frame(  
  Product = rownames(sales),  
  January = sales[,1],
```

```

    February = sales[,2],
    March = sales[,3]
  )
  ui <- fluidPage(
    titlePanel("Product Sales Dashboard"),
    fluidRow(
      column(6,
        plotOutput("barPlot")),
      column(6,
        plotOutput("areaPlot"))
    ),
    hr(),
    tableOutput("salesTable")
  )
  server <- function(input, output) {
    output$barPlot <- renderPlot({
      barplot(sales,
        beside=TRUE,
        col=c("red", "blue", "green"),
        main="Grouped Bar Chart",
        xlab="Months",
        ylab="Sales")
      legend("topright",
        legend=rownames(sales),
        fill=c("red", "blue", "green"))
    })
    output$areaPlot <- renderPlot({
      months <- 1:3
      plot(months, sales[1,],
        type="n",
        ylim=c(0,6000),
        xaxt="n",
        xlab="Months",
        ylab="Sales",
        main="Stacked Area Chart")
      axis(1, at=1:3, labels=c("Jan", "Feb", "Mar"))
      polygon(c(months, rev(months)),
        c(rep(0,3), rev(sales[1,])),
        col="red")
    })
  }
}

```

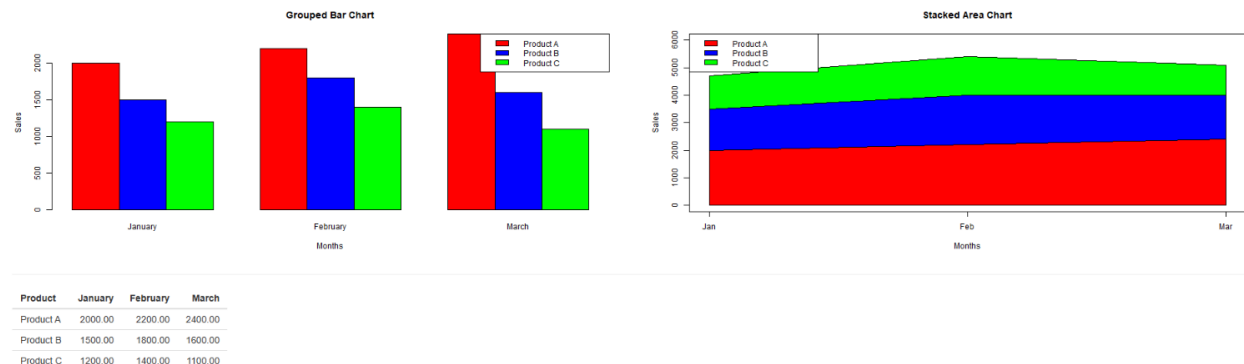
```

polygon(c(months,rev(months)),
       c(sales[1,],rev(sales[1,]+sales[2,])),
       col="blue")
polygon(c(months,rev(months)),
       c(sales[1,]+sales[2,],
         rev(sales[1,]+sales[2,]+sales[3,])),
       col="green")
legend("topleft",
      legend=rownames(sales),
      fill=c("red", "blue", "green"))
})
output$salesTable <- renderTable({
  sales_table
})
}
shinyApp(ui, server)

```

Output:

Product Sales Dashboard



Set 7 Customer Demographics Analysis

1: Bar Chart

Code:

```

customers <- data.frame(
  CustomerID=c(1,2,3),
  Age=c(28,35,42),
  Gender=c("Female","Male","Female"),
  Income=c(50000,60000,75000)
)
barplot(customers$Age,
      names.arg=customers$CustomerID,
      col="skyblue",

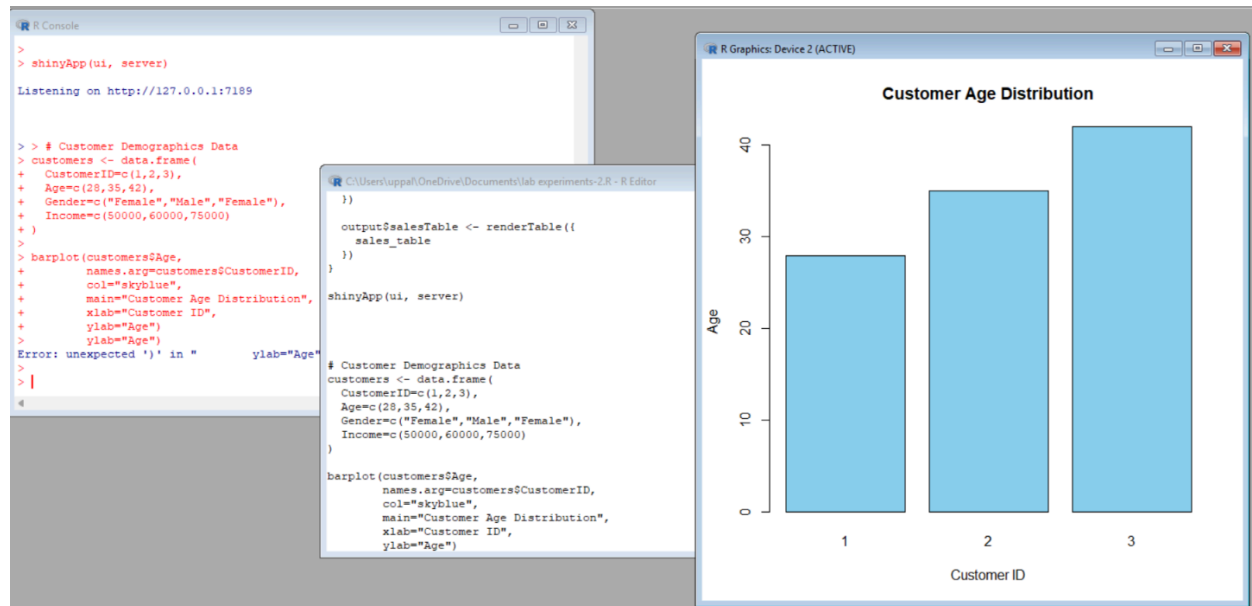
```

```

main="Customer Age Distribution",
xlab="Customer ID",
ylab="Age")

```

Output:



2: Pie Chart

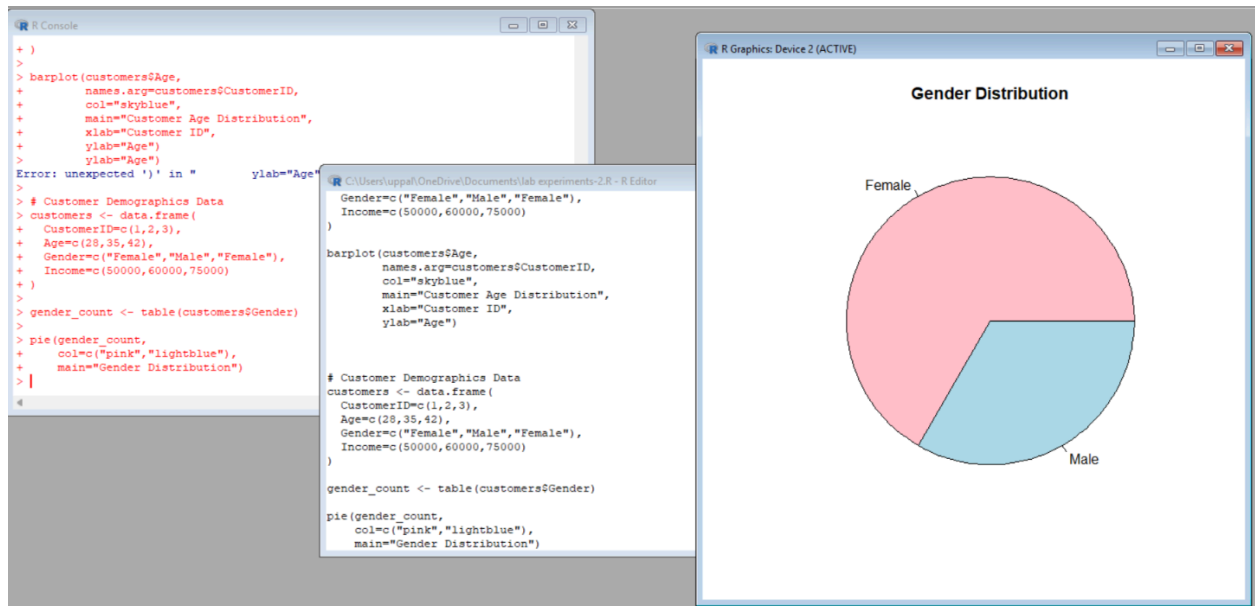
Code:

```

customers <- data.frame(
  CustomerID=c(1,2,3),
  Age=c(28,35,42),
  Gender=c("Female","Male","Female"),
  Income=c(50000,60000,75000)
)
gender_count <- table(customers$Gender)
pie(gender_count,
  col=c("pink","lightblue"),
  main="Gender Distribution")

```

Output:



3: Table

Code:

```

customers <- data.frame(
  CustomerID=c(1,2,3),
  Age=c(28,35,42),
  Gender=c("Female","Male","Female"),
  Income=c(50000,60000,75000)
)

```

```

print(customers)

```

Output:

```

>
> pie(gender_count,
+     col=c("pink","lightblue"),
+     main="Gender Distribution")
> customers <- data.frame(
+   CustomerID=c(1,2,3),
+   Age=c(28,35,42),
+   Gender=c("Female","Male","Female"),
+   Income=c(50000,60000,75000)
+ )
>
> print(customers)
  CustomerID Age Gender Income
1          1  28 Female 50000
2          2  35  Male 60000
3          3  42 Female 75000
>

```

```

C:\Users\uppa\OneDrive\Documents\lab experiments-2.R - R Editor

      ylab="Age")

# Customer Demographics Data
customers <- data.frame(
  CustomerID=c(1,2,3),
  Age=c(28,35,42),
  Gender=c("Female","Male","Female"),
  Income=c(50000,60000,75000)
)

gender_count <- table(customers$Gender)

pie(gender_count,
    col=c("pink","lightblue"),
    main="Gender Distribution")

customers <- data.frame(
  CustomerID=c(1,2,3),
  Age=c(28,35,42),
  Gender=c("Female","Male","Female"),
  Income=c(50000,60000,75000)
)

print(customers)

```

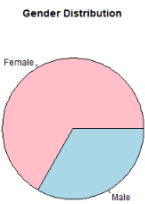
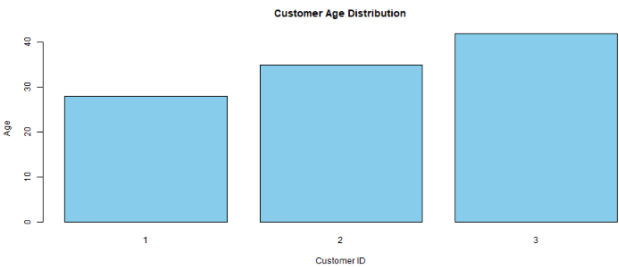
4: Dashboard (R Shiny)

Code:

```
library(shiny)
customers <- data.frame(
  CustomerID=c(1,2,3),
  Age=c(28,35,42),
  Gender=c("Female","Male","Female"),
  Income=c(50000,60000,75000)
)
ui <- fluidPage(
  titlePanel("Customer Demographics Dashboard"),
  fluidRow(
    column(6, plotOutput("bar")),
    column(6, plotOutput("pie"))
  ),
  hr(),
  tableOutput("table")
)
server <- function(input, output){
  output$bar <- renderPlot({
    barplot(customers$Age,
      names.arg=customers$CustomerID,
      col="skyblue",
      main="Customer Age Distribution",
      xlab="Customer ID",
      ylab="Age")
  })
  output$pie <- renderPlot({
    pie(table(customers$Gender),
      col=c("pink","lightblue"),
      main="Gender Distribution")
  })
  output$table <- renderTable({
    customers
  })
}
shinyApp(ui, server)
```

Output:

Customer Demographics Dashboard



CustomerID	Age	Gender	Income
1.00	28.00	Female	50000.00
2.00	35.00	Male	60000.00
3.00	42.00	Female	75000.00

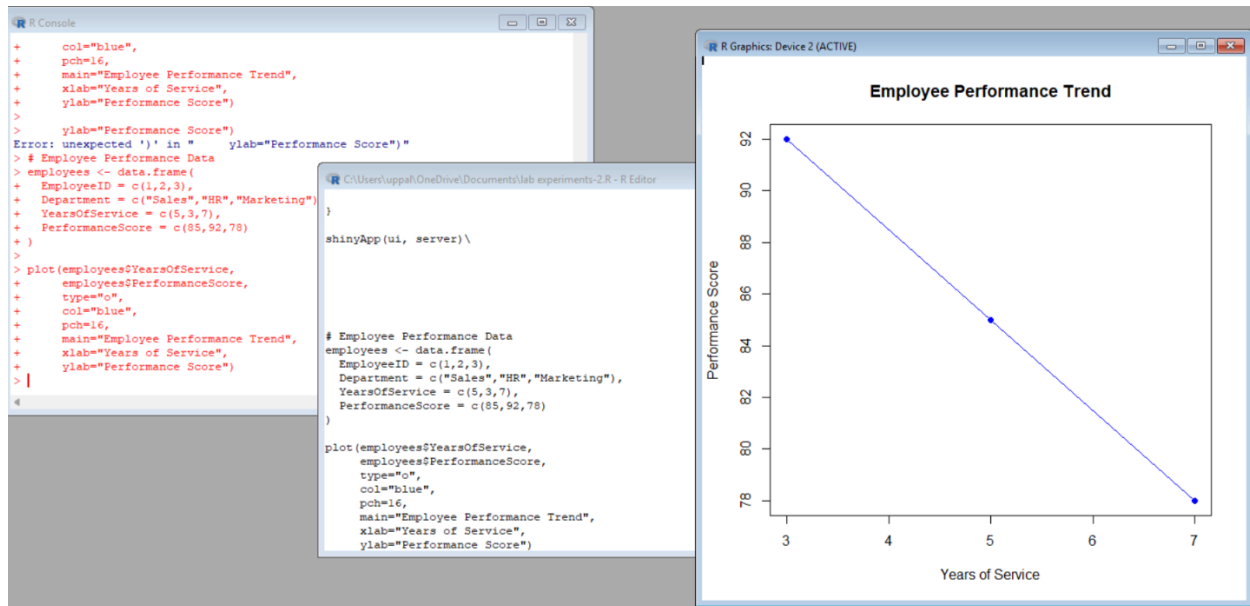
Set 8 Employee Performance Analysis

1: Line Chart

Code:

```
employees <- data.frame(  
  EmployeeID = c(1,2,3),  
  Department = c("Sales","HR","Marketing"),  
  YearsOfService = c(5,3,7),  
  PerformanceScore = c(85,92,78)  
)  
plot(employees$YearsOfService,  
     employees$PerformanceScore,  
     type="o",  
     col="blue",  
     pch=16,  
     main="Employee Performance Trend",  
     xlab="Years of Service",  
     ylab="Performance Score")
```

Output:



2: Bar Chart

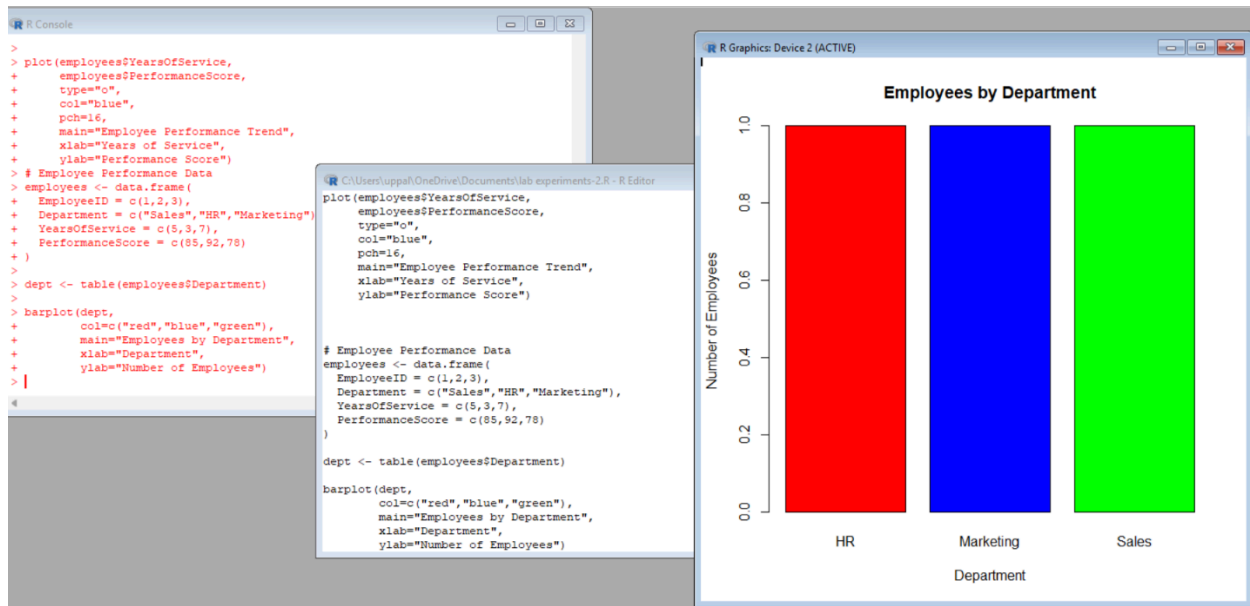
Code:

```

employees <- data.frame(
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsOfService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)
dept <- table(employees$Department)
barplot(dept,
  col=c("red","blue","green"),
  main="Employees by Department",
  xlab="Department",
  ylab="Number of Employees")

```

Output:

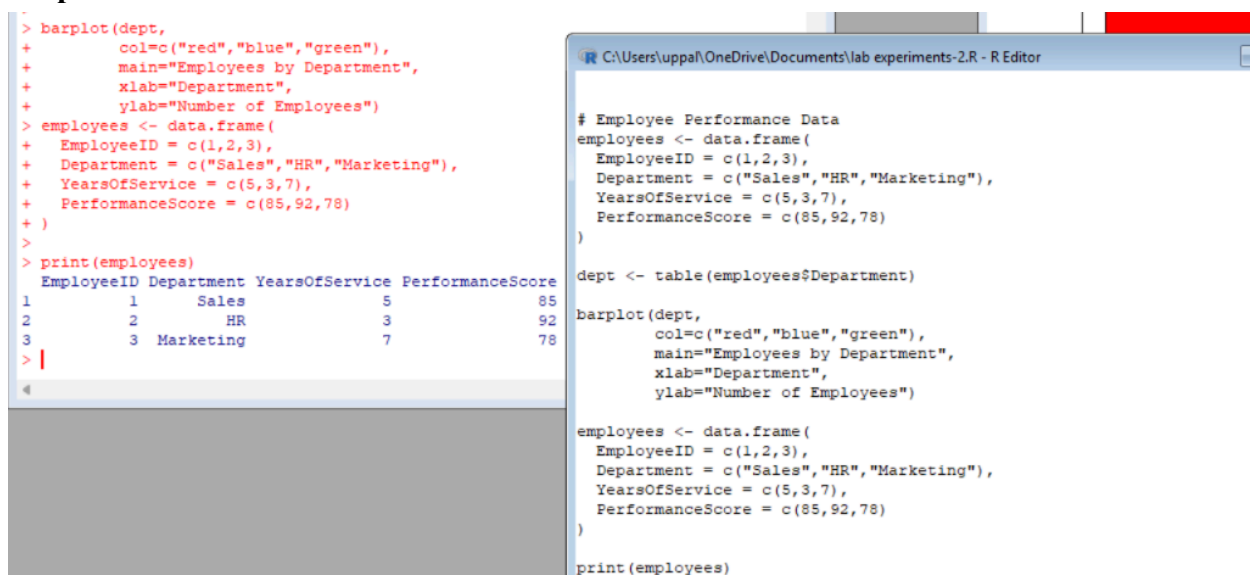


3: Table

Code:

```
employees <- data.frame(
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsOfService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)
print(employees)
```

Output:



4: Dashboard (R Shiny)

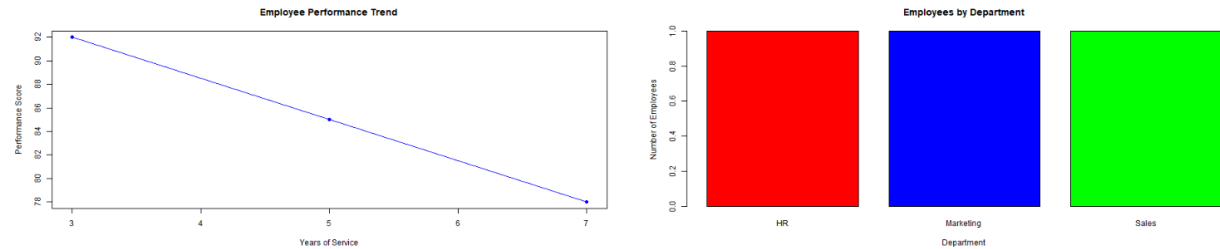
Code:

```
library(shiny)
employees <- data.frame(
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsOfService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)
ui <- fluidPage(
  titlePanel("Employee Performance Dashboard"),
  fluidRow(
    column(6, plotOutput("linePlot")),
    column(6, plotOutput("barPlot"))
  ),
  hr(),
  tableOutput("empTable")
)
server <- function(input, output){
  output$linePlot <- renderPlot({
    plot(employees$YearsOfService,
         employees$PerformanceScore,
         type="o",
         col="blue",
         pch=16,
         main="Employee Performance Trend",
         xlab="Years of Service",
         ylab="Performance Score")
  })
  output$barPlot <- renderPlot({
    dept <- table(employees$Department)
    barplot(dept,
            col=c("red","blue","green"),
            main="Employees by Department",
            xlab="Department",
            ylab="Number of Employees")
  })
  output$empTable <- renderTable({
    employees
  })
}
```

```
shinyApp(ui, server)
```

Output:

Employee Performance Dashboard



EmployeeID	Department	YearsOfService	PerformanceScore
1.00	Sales	5.00	85.00
2.00	HR	3.00	92.00
3.00	Marketing	7.00	78.00

set 9. Online Learning Activity

1: Histogram and Boxplot

Code:

```
student <- data.frame(  
  Student_ID = c("L01", "L02", "L03", "L04", "L05", "L06"),  
  Gender = c("Male", "Female", "Male", "Female", "Male", "Female"),  
  Age = c(20, 22, 19, 21, 23, 20),  
  Course = c("R", "R", "SQL", "R", "R", "SQL"),  
  Study_Time = c(3.5, 4.2, 2.0, 5.0, 2.5, 4.0),  
  Videos_Watched = c(12, 15, 8, 18, 9, 14),  
  Quiz_Score = c(78, 85, 65, 92, 70, 88),  
  Login_Date = c("2025-01-05", "2025-01-05", "2025-02-08",  
    "2025-02-08", "2025-03-12", "2025-03-12"),  
  stringsAsFactors = FALSE  
)  
hist(student$Quiz_Score,  
  col="skyblue",  
  main="Histogram of Quiz Scores",  
  xlab="Quiz Score",  
  ylab="Frequency")  
boxplot(student$Quiz_Score ~ student$Course,  
  col=c("pink", "lightgreen"),  
  main="Quiz Score by Course",  
  xlab="Course",  
  ylab="Quiz Score")
```

Output:

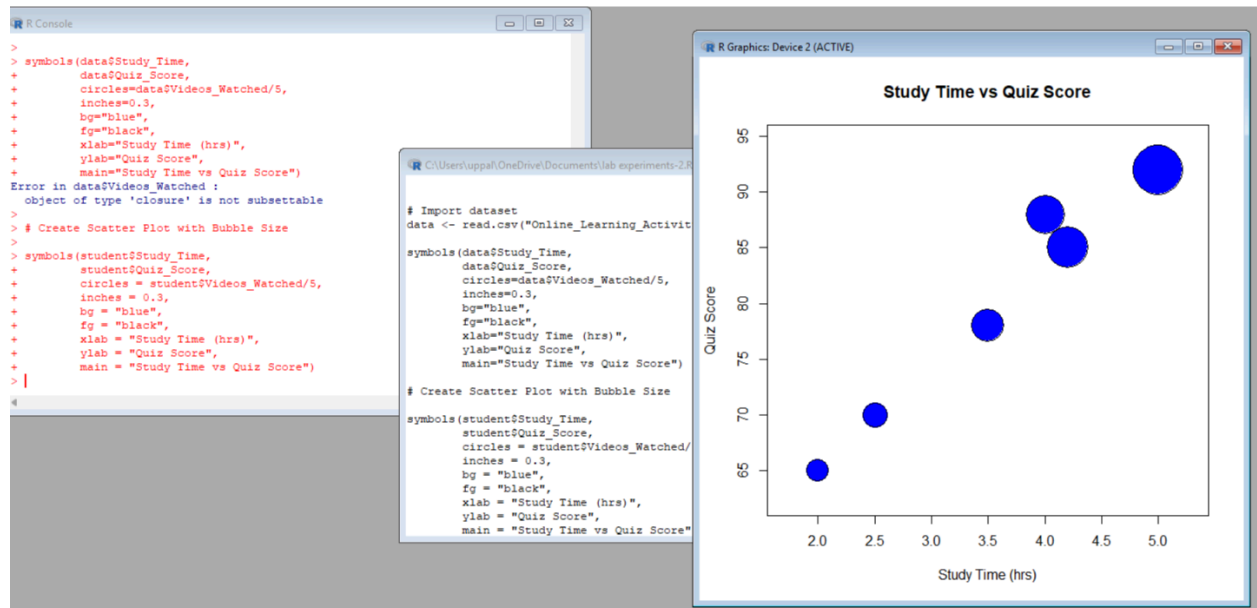


2: Scatter Plot with Bubble Size

Code:

```
data <- read.csv("Online_Learning_Activity.csv")  
symbols(data$Study_Time,  
        data$Quiz_Score,  
        circles=data$Videos_Watched/5,  
        inches=0.3,  
        bg="blue",  
        fg="black",  
        xlab="Study Time (hrs)",  
        ylab="Quiz Score",  
        main="Study Time vs Quiz Score")
```

Output:



3: Date Conversion and Line Chart

Code:

```

student$Login_Date <- as.Date(student$Login_Date)
student$Month <- format(student$Login_Date, "%Y-%m")
avg_score <- aggregate(Quiz_Score ~ Month, data = student, mean)
print(avg_score)
plot(avg_score$Quiz_Score,
     type="o",
     xaxt="n",
     col="blue",
     lwd=2,
     pch=16,
     xlab="Month",
     ylab="Average Quiz Score",
     main="Average Monthly Quiz Score")

axis(1,
     at=1:nrow(avg_score),
     labels=avg_score$Month)
ma <- stats::filter(avg_score$Quiz_Score,
                    rep(1/2,2),
                    sides=1)

lines(ma,
      col="red",
      lwd=2)

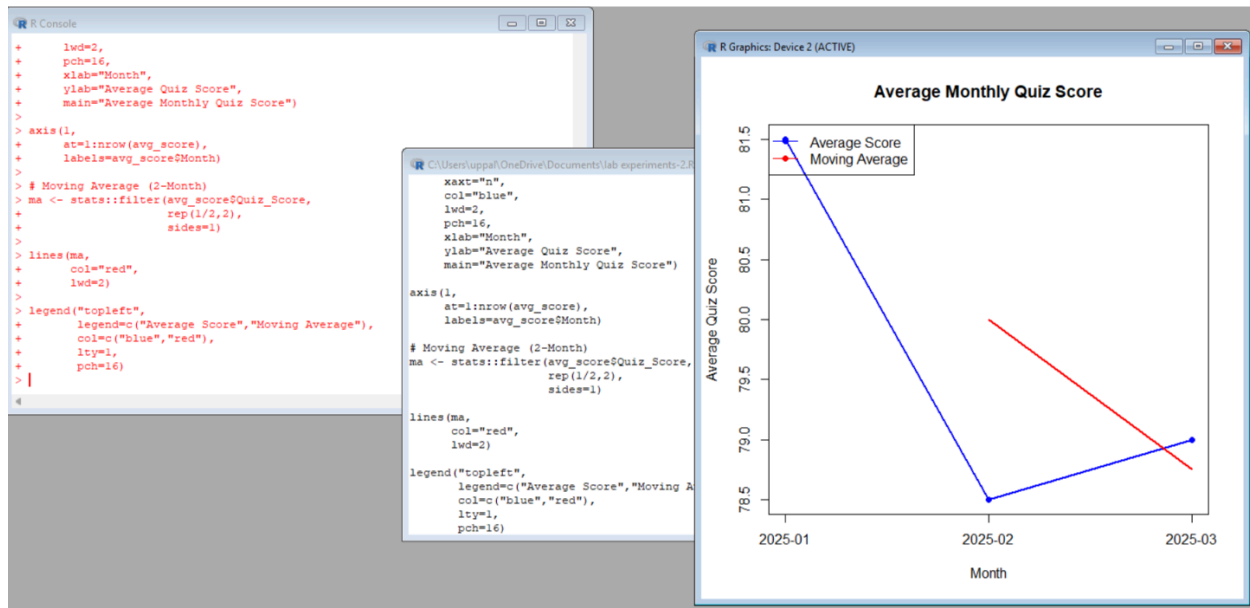
```

```

legend("topleft",
      legend=c("Average Score","Moving Average"),
      col=c("blue","red"),
      lty=1,
      pch=16)

```

Output:



4: Dashboard (R Shiny)

Code:

```

library(shiny)
student <- data.frame(
  Student_ID = c("L01","L02","L03","L04","L05","L06"),
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Age = c(20,22,19,21,23,20),
  Course = c("R","R","SQL","R","R","SQL"),
  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),
  Videos_Watched = c(12,15,8,18,9,14),
  Quiz_Score = c(78,85,65,92,70,88)
)
ui <- fluidPage(
  titlePanel("Online Learning Activity Dashboard"),
  sidebarLayout(
    sidebarPanel(
      selectInput("gender",
                  "Select Gender",

```



```

        choices = c("All", unique(student$Gender)),
        selected = "All"),
    selectInput("course",
        "Select Course",
        choices = c("All", unique(student$Course)), selected = "All")
),
mainPanel(
    fluidRow(
        column(6, plotOutput("barPlot")),
        column(6, plotOutput("scatterPlot"))
    ),
    br(),
    tableOutput("table")

)
)
)
server <- function(input, output){
    filtered <- reactive({
        df <- student
        if(input$gender != "All")
            df <- df[df$Gender == input$gender, ]

        if(input$course != "All")
            df <- df[df$Course == input$course, ]
        df
    })
    output$barPlot <- renderPlot({
        avg <- aggregate(Quiz_Score ~ Course,
            filtered(),
            mean)
        barplot(avg$Quiz_Score,
            names.arg = avg$Course,
            col = c("orange", "green"),
            main = "Average Quiz Score by Course",
            xlab = "Course",
            ylab = "Average Quiz Score")
    })
    output$scatterPlot <- renderPlot({
        df <- filtered()

```

```

symbols(df$Study_Time,
        df$Quiz_Score,
        circles = df$Videos_Watched/5,
        inches = 0.3,
        bg = "blue",
        fg = "black",
        xlab = "Study Time (hrs)",
        ylab = "Quiz Score",
        main = "Study Time vs Quiz Score")
})
output$table <- renderTable({
  filtered()
})
}
shinyApp(ui, server)

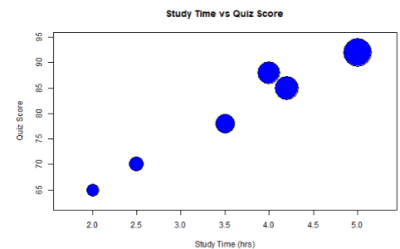
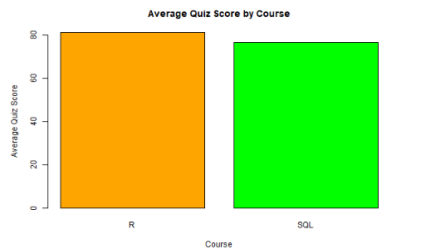
```

Output:

Online Learning Activity Dashboard

Select Gender
All

Select Course
All



Student_ID	Gender	Age	Course	Study_Time	Videos_Watched	Quiz_Score
L01	Male	20.00	R	3.50	12.00	76.00
L02	Female	22.00	R	4.20	15.00	85.00
L03	Male	19.00	SQL	2.00	8.00	65.00
L04	Female	21.00	R	5.00	18.00	92.00
L05	Male	23.00	R	2.50	9.00	70.00
L06	Female	20.00	SQL	4.00	14.00	88.00

Set 10 Survey Responses Analysis

1: Grouped Bar Chart

Code:

```

survey <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)
q1 <- table(survey$Question1)
barplot(q1,

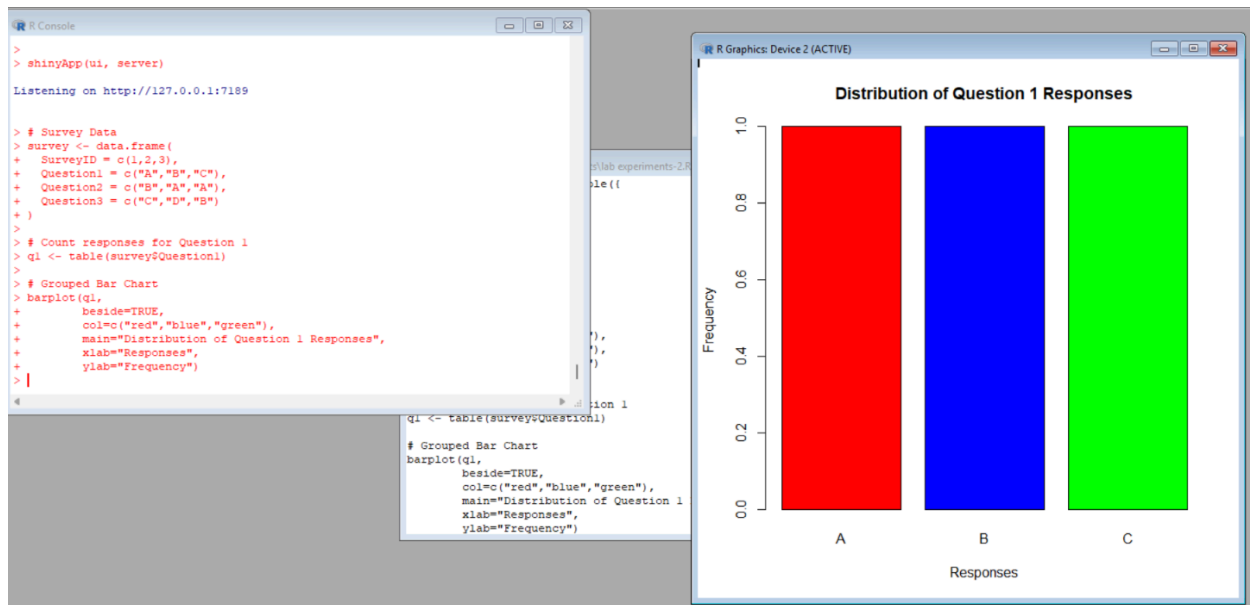
```

```

beside=TRUE,
col=c("red","blue","green"),
main="Distribution of Question 1 Responses",
xlab="Responses",
ylab="Frequency")

```

Output:



2: Stacked Bar Chart

Code:

```

survey <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)
responses <- rbind(
  table(factor(survey$Question1, levels=c("A","B","C","D"))),
  table(factor(survey$Question2, levels=c("A","B","C","D"))),
  table(factor(survey$Question3, levels=c("A","B","C","D")))
)
rownames(responses) <- c("Question1","Question2","Question3")
barplot(t(responses),
  col=c("red","blue","green","yellow"),
  main="Overall Survey Responses",
  xlab="Questions",
  ylab="Frequency")

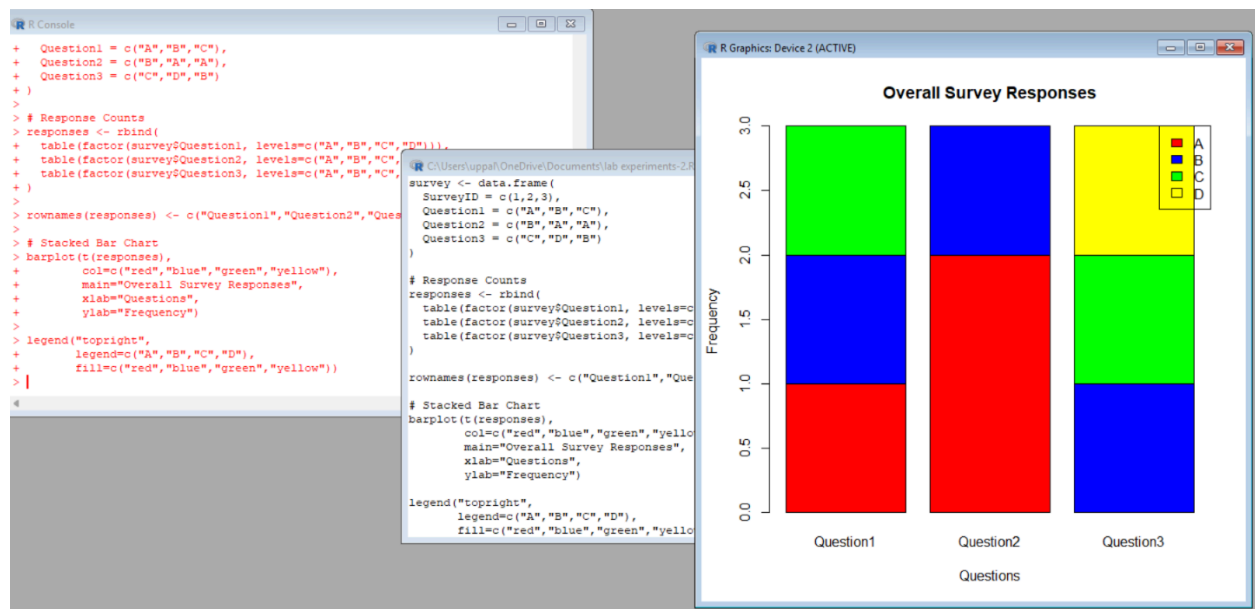
```

```

legend("topright",
      legend=c("A","B","C","D"),
      fill=c("red","blue","green","yellow"))

```

Output:



3: Table

Code:

```

survey <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)
print(survey)

```

Output:

```

> legend("topright",
+       legend=c("A","B","C","D"),
+       fill=c("red","blue","green","yellow"))
> survey <- data.frame(
+   SurveyID = c(1,2,3),
+   Question1 = c("A","B","C"),
+   Question2 = c("B","A","A"),
+   Question3 = c("C","D","B")
+ )
> print(survey)
  SurveyID Question1 Question2 Question3
1         1         A         B         C
2         2         B         A         D
3         3         C         A         B

```

```

R C:\Users\uppal\OneDrive\Documents\lab experiments-2.R - R Editor
table(factor(survey$Question1, levels=c("A","B","C","D"))),
table(factor(survey$Question2, levels=c("A","B","C","D"))),
table(factor(survey$Question3, levels=c("A","B","C","D")))
)

rownames(responses) <- c("Question1","Question2","Question3")

# Stacked Bar Chart
barplot(t(responses),
        col=c("red","blue","green","yellow"),
        main="Overall Survey Responses",
        xlab="Questions",
        ylab="Frequency")

legend("topright",
       legend=c("A","B","C","D"),
       fill=c("red","blue","green","yellow"))

survey <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)

print(survey)

```

4: Dashboard (R Shiny)

Code:

```

library(shiny)
survey <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)
ui <- fluidPage(
  titlePanel("Survey Responses Dashboard"),
  fluidRow(
    column(6, plotOutput("groupBar")),
    column(6, plotOutput("stackBar"))
  ),
  br(),
  tableOutput("surveyTable")
)
server <- function(input, output){
  output$groupBar <- renderPlot({
    q1 <- table(survey$Question1)
    barplot(q1,
            beside=TRUE,
            col=c("red","blue","green"),
            main="Question 1 Responses",
            xlab="Responses",

```

```

      ylab="Frequency")
  })
  output$stackBar <- renderPlot({
    responses <- rbind(
      table(factor(survey$Question1, levels=c("A","B","C","D"))),
      table(factor(survey$Question2, levels=c("A","B","C","D"))),
      table(factor(survey$Question3, levels=c("A","B","C","D")))
    )
    rownames(responses) <- c("Question1","Question2","Question3")
    barplot(t(responses),
      col=c("red","blue","green","yellow"),
      main="Overall Survey Responses",
      xlab="Questions",
      ylab="Frequency")
    legend("topright",
      legend=c("A","B","C","D"),
      fill=c("red","blue","green","yellow"))
  })
  output$surveyTable <- renderTable({
    survey
  })
}
shinyApp(ui, server)

```

Output:

Survey Responses Dashboard

